

Aside from orbits, we also observe the model encoding particular key tokens as “sliders”, as seen in the middle of the bottom row in Figure 12 (which is the token “wrong”, from the same message as already shown in Figure 11). In these motions the trajectory noticeably drifts in a single direction, which the model could use to implement a mechanism to count how many iterations have occurred.

The emergence of structured trajectories in latent space gives us a glimpse into how the model performs its computations. Unlike the discrete sequential chain of reasoning seen in verbalized chain-of-thought approaches, we observe rich geometric patterns including orbits, convergent paths, and drifts - means to organize its computational process spatially. This suggests the model is independently learning to leverage the high-dimensional nature of its latent space to implement reasoning in new ways.

Path Independence. We verify that our models maintain path independence, in the sense of Anil et al. (2022), despite their complex, learned dynamics, which we discussed prior (see also the additional examples in Appendix Figure 22). When re-initializing from multiple starting states s_0 , the model moves in similar trajectories, exhibiting consistent behavior. The same orbital patterns, fixed points, or directional drifts emerge regardless of initialization.

8. Related Work Overview

The extent to which recurrence is a foundational concept of machine learning is hard to overstate (Amari, 1972; Hopfield, 1982; Braitenberg, 1986; Gers and Schmidhuber, 2000; Sutskever et al., 2008). Aside from using recurrence to move along sequences, as in recurrent neural networks, it was understood early to also be the key to adaptive computation (Schmidhuber, 2012; Graves, 2017). For transformers, recurrence was applied in Dehghani et al. (2019), who highlight the aim of recurrent depth to model *universal*, i.e. Turing-complete, machines (Graves et al., 2014). It was used at scale (but with fixed recurrence) in Lan et al. (2019) and an interesting recent improvement in this line of work are described in Tan et al. (2023); Abnar et al. (2023), Mathur et al. (2024) and Csordás et al. (2024). Schwarzschild et al. (2021b); Bansal et al. (2022); Bear et al. (2024) and McLeish et al. (2024) show that depth recurrence is advantageous when learning generalizable algorithms when training with randomized unrolling and input injections. Recent work has described depth-recurrent, *looped*, transformers and studied their potential benefits with careful theoretical and small-scale analysis (Giannou et al., 2023; Gatmiry et al., 2024; Yang et al., 2024a; Fan et al., 2025).

From another angle, these models can be described as neural networks learning a fixed-point iteration, as studied in *deep equilibrium* models (Bai et al., 2019; 2022). They

are further related to diffusion models (Song and Ermon, 2019), especially latent diffusion models (Rombach et al., 2022), but we note that language diffusion models are usually run with a per-sequence, instead of a per-token, iteration count (Lee et al., 2018). A key difference of our approach to both equilibrium models and diffusion models is in the training objective, where equilibrium methods solve the “direct” problem (Geiping and Moeller, 2019), diffusion models solve a surrogate training objective, and our work suggests that truncated unrolling is a scalable alternative.

More generally, all architectures that recur in depth can also be understood as directly learning the analog to the gradient of a latent energy-based model (LeCun and Huang, 2005; LeCun, 2022), to an implicitly defined intermediate optimization layer (Amos and Kolter, 2017), or to a Kuramoto layer (Miyato et al., 2024). Analogies to gradient descent at inference time also show the connection to test time adaptation (Sun et al., 2020), especially test-time adaptation of output states (Boudiaf et al., 2022).

Aside from full recurrent-depth architectures, there also exist a number of proposals for hybrid architectures, such as models with latent sub-networks (Li et al., 2020a), LoRA adapters on top of weight-shared layers (Bae et al., 2024), or (dynamic) weight-tying of trained models (Hay and Wolf, 2023; Liu et al., 2024b).

As mentioned in Section 6, while we consider the proposed recurrent depth approach to be a very natural way to learn to reason in continuous latent space from the ground up, the works of Hao et al. (2024); Cheng and Durme (2024) and Liu et al. (2024a) discuss how to finetune existing fixed-depth transformers with this capability. These works have a similar aim to ours, enabling reasoning in latent space, but approach this goal from separate directions.

For additional discussions related to the idea of constructing a prior that incentivizes reasoning and algorithm learning at the expense of memorization of simple patterns, we also refer to Chollet (2019), Schwarzschild (2023), Li et al. (2020b) and Moulton (2023).

9. Future Work

Aside from work extending and analyzing the scaling behaviors of recurrent depth models, there are many questions that remain unanswered. For example, to us, there are potentially a large number of novel post-training schemes that further enhance the capabilities of these models, such as fine-tuning to compress the recurrence or reinforcement learning with data with different hardness levels (Zelikman et al., 2024), or to internalize reasoning from CoT data into the recurrence (Deng et al., 2024).

Another aspect not covered in this work is the relationship

to other modern architecture improvements. Efficient sequence mixing operations, especially those that are linear in sequence dimension, such as linear attention (Katharopoulos et al., 2020; Yang et al., 2024b), are limited in the number of comparisons that can be made. However, with recurrent depth, blocks containing linear operators can repeat until all necessary comparisons between sequence elements are computed (Suzgun et al., 2019). For simplicity, we also focus on a single recurrence, where prior work has considered multiple successive recurrent stages (Takase and Kiyono, 2023; Csordás et al., 2024).

Finally, the proposed architecture is set up to be *compute-heavy*, with more “materialized” parameters than there are actual parameters. This naturally mirrors mixture-of-expert models (MoE), which are *parameter-heavy*, using fewer active parameters per forward pass than exist within the model (Shazeer et al., 2017; Fedus et al., 2022). We posit that where the recurrent-depth setup excels at learning reasoning patterns, the MoE excels at effectively storing and retrieving complex information. Their complementarity supports the hypothesis that a future architecture would contain both modifications. While in a standard MoE model, each expert can only be activated once per forward pass, or skipped entirely, a recurrent MoE model could also refine its latent state over multiple iterations, potentially routing to the same expert multiple times, before switching to a different one (Tan et al., 2023; Csordás et al., 2024). While MoE models are the currently leading solution to implement this type of “memory” in dense transformers, these considerations also hold for other memory mechanisms suggested for LLMs (Sukhbaatar et al., 2019; Fan et al., 2021; Wu et al., 2022; He et al., 2024).

10. Conclusions

The models described in this paper are ultimately still a proof-of-concept. We describe how to train a latent recurrent-depth architecture, what parameters we chose, and then trained a single model at scale. Future training runs are likely to train with more optimized learning rate schedules, data mixes and accelerators. Still we observe a number of interesting behaviors emerging naturally from recurrent training. The most important of these is the ability to use latent reasoning to dramatically improve performance on reasoning tasks by expending test-time computation. In addition, we also observe context-dependent convergence speed, path independence, and various zero-shot abilities. This leads us to believe that latent reasoning is a promising research direction to complement existing approaches for test-time compute scaling. The model we realize is surprisingly powerful given its size and amount of training data, and we are excited about the potential impact of imbuing generative models with the ability to reason in continuous

latent space without the need for specialized data at train time or verbalization at inference time.

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